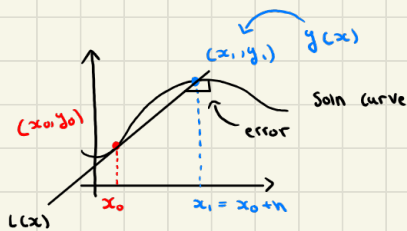


Numerical Methods



Where h is the distance between consecutive points in your numerical. It is not the distance from x_0 to x_1 .

Approximate $y(x_1)$ using a tangent line $h \rightarrow$ step size w
usually a small number, $h = 0.1, 0.2, 0.5$ $h \cdot n$ is

Euler's method:

The procedure, we use the linearization of the unknown soln, $y(x)$ of $y' = f(x, y)$, $y(x_0) = y_0$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$x_n = x_0 + nh, \quad n = 0, 1, 2, \dots$$

Euler's method

• when you are using Euler's method the $f(x_n, y_n)$ should be given to you with the derivative already taken

Practice Problems

Given $y = x^2$, $h = 0.1$, $P(1,1)$ Find $y(1.5)$

n	x_n	y_n
0	1	1
1	1.1	1.2
2	1.2	1.42
3	1.3	1.66
4	1.4	1.92
5	1.5	2.20

★ always begin with $n=0$ the $(1,1)$ were given

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{0+1} = y_0 + h f(x_0, y_0)$$

$$y_1 = y_0 + h f(x_0, y_0)$$

keep add h value until you get to your favored amount

all you now have to do is look at the table and plug numbers into the equation looking at the table $y_0 = 1$ and $x_0 = 1$, $y_0 = 1$

$$y_1 = 1 + 0.1 f(1, 1)$$

$$y_1 = 1 + 0.1 [2x]$$

$$y_1 = 1 + 0.1 [2(1)]$$

$$y_1 = 1.2$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{1+1} = y_1 + h f(x_1, y_1)$$

$$y_2 = y_1 + h [2x_1]$$

$$y_2 = 1.2 + 0.1 [2(1.1)]$$

$$y_2 = 1.42$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{2+1} = y_2 + h f(x_2, y_2)$$

$$y_3 = y_2 + h f(x_2, y_2)$$

$$y_3 = (1.42) + (0.1) [2x_2]$$

$$y_3 = (1.42) + (0.1) [2(1.2)]$$

$$y_3 = 1.66$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{3+1} = y_3 + h f(x_3, y_3)$$

$$y_4 = y_3 + h f(x_3, y_3)$$

$$y_4 = (1.66) + (0.1) [2(x_3)]$$

$$y_4 = (1.66) + 0.1 [2(1.3)]$$

$$y_4 = 1.92$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{4+1} = y_4 + h f(x_4, y_4)$$

$$y_5 = y_4 + h f(x_4, y_4)$$

$$y_5 = (1.92) + 0.1 [2x_4]$$

$$y_5 = (1.92) + 0.1 [2(1.4)]$$

$$y_5 = 2.20$$

$$y(1.5) = 2.20$$

$$y' = 2x - 3y + 1, \quad y(1) = 6 \quad y(1.2) = ? \quad , \quad h = 0.05$$

n	x_n	y_n
0	1	6
1	1.05	5.25
2	1.1	4.6175
3	1.15	4.084875
4	1.2	3.6371

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{0+1} = y_0 + h f(x_0, y_0)$$

$$y_1 = 6 + 0.05 [2(1) - 3(6) + 1]$$

$$y_1 = 5.25$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{1+1} = y_1 + h f(x_1, y_1)$$

$$y_2 = 5.25 + (0.05) [2(1.05) - 3(5.25) + 1]$$

$$y_2 = 4.6175$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{2+1} = y_2 + h f(x_2, y_2)$$

$$y_3 = (4.6175) + (0.05) [2(1.1) - 3(4.6175) + 1]$$

$$y_3 = 4.084875$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{3+1} = y_3 + h f(x_3, y_3)$$

$$y_4 = 4.084875 + (0.05) [2(1.15) - 3(4.084875) + 1]$$

$$y_4 = 3.6371$$

$$y(1.2) = 3.6371$$

✱ wait until the end to round your answer do not round on individual steps round answer to four decimal places